07 · Huntington's disease

The words of this track, as it says them. Bold brown is the examiner.

Huntington's disease. Five marks.

Explain Huntington's disease as a genetic disease: its cause, how it damages the body, how it's inherited, and the chance of an affected child when one parent is big H little h, and the other is little h little h.

It's on her help board for Module 3, and it comes back in the final, in case 15. So learn it once, and use it twice.

The hook: chromosome four, a stutter in the gene, a toxic protein in the brain. And one big H is enough.

The cause. Huntington's is caused by a mutated gene on chromosome four, and chromosome four is an autosome. Her help board puts it in numbers: humans have forty-six chromosomes, forty-four autosomes and two sex chromosomes.

What the mutation is. A copying error during DNA replication, for example while sperm are being made. The base sequence C A G is repeated too many times: thirty-six to a hundred and twenty times, instead of the normal ten to thirty-five. Think of it as the gene stuttering.

How it damages the body. The longer gene makes a larger, abnormal protein called huntingtin. It's toxic. It builds up in the brain and damages brain cells, especially the ones for movement, thinking and memory.

How it's inherited. Autosomal dominant. One copy of the mutated allele, big H little h, is enough to cause the disease, and it can come from either parent. That means there are no healthy carriers. Anyone with a big H develops the disease, and only little h little h is unaffected. It shows in every generation.

The cross. Big H little h, crossed with little h little h. Two boxes are big H little h, and two are little h little h. Each child has a fifty per cent chance of being affected.

The traps. Dominant means big H little h is affected, not a carrier. Her deck asks it straight out: can a person be a healthy carrier? No. Second: chromosome four is an autosome, not a sex chromosome. Third: it's a gene mutation, a repeat, not an extra or missing chromosome, so it isn't aneuploidy.

Once more. Huntington's: the cause, the damage, the inheritance, and the chance.

A mutated gene on chromosome four, an autosome: C A G repeated thirty-six to a hundred and twenty times, instead of ten to thirty-five. It makes an abnormal, toxic huntingtin protein that builds up in the brain and damages movement, thinking and memory. It's autosomal dominant: one copy is enough, there are no healthy carriers, and it shows in every generation. Big H little h, crossed with little h little h, gives a fifty per cent chance for each child.